

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	22 june 2025
Team ID	LTVIP2025TMID52313
Project Name	Measuring the Pulse of Prosperity: An Index of Economic Freedom Analysis
Maximum Marks	4 Marks

Technical Architecture:

The system collects structured datasets (e.g. CSV/Excel) on economic freedom, stores them in a relational database (MySQL), and presents them through a web-based interactive dashboard built with Tableau. Data is processed and cleaned using Python for visualization purposes.

Infrastructure: Hybrid (Local for development, Cloud for visualization hosting if applicable)

External Interfaces: Economic data sources (Heritage Index, IMF APIs if used)

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web-based interactive dashboards	Tableau
2	Application Logic-1	Data pre-processing & EDA logic	Python (Pandas, NumPy, Matplotlib)
3	Application Logic-2	Data transformation & connection to Tableau	Tableau Prep / Python
4	Database	Relational database for storing economic data	MySQL
5	Cloud Database	Optionally used if deployed in cloud	Amazon RDS / Google Cloud SQL
6	File Storage	CSV/Excel files of economic data	Local Storage / Cloud Drive
7	External API-1	Used for fetching global economic data (optional)	IMF API / Heritage API
8	Infrastructure	Deployment environment	Local (Dev), Cloud (Deployment via Tableau Public / Server)

Table-2: Application Characteristics:

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	Python packages used for data cleaning & visualization	Pandas, NumPy, Seaborn
2	Security Implementations	Securing access to dashboards & data exports	Tableau Access Permissions, Authentication via Email
3	Scalable Architecture	Dashboards & database can scale horizontally	Tableau Server / Tableau Cloud, MySQL Scaling
4	Availability	Hosted dashboards are accessible 24/7 online	Tableau Public / Cloud-hosted reports
5	Performance	Quick dashboard load, filter optimization, efficient querying	Tableau Extracts, Indexed MySQL Tables